
A geometric relation of the error introduced by sampling a language model’s output distribution to its internal state

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Abstract

GPT-style language models are sensitive to single-token changes at generation points where the predicted probability distribution is spread across multiple tokens. Viewing this sensitivity as a geometric property, we derive an $\mathfrak{so}(n)$ -valued 1-form that depends only on the geometry of the token embeddings. Despite this purely geometric origin, we show that its curvature is semantically meaningful: On chess reasoning tasks, the curvature couples to the world model of an off-the-shelf instruction-tuned model, with transformations clustering by board region and respecting piece importance. Our findings suggest that token space geometry directly reflects how models internally represent problems.

1. Introduction

One can generate simple images by training a feed-forward neural network to map one-hot encoded labels to a $w \times h$: $w, h \in \mathbb{N}$ sized vector of pixel brightness values. It is well known that such a simple architecture’s reproductions will only ever be blurry, as the best solution to the given mean-square-error optimization problem is to reproduce the average image associated with each label.

Let $\mathcal{Z} \cong \mathbb{R}^n$ be the vector space of a GPT-style (Radford et al., 2018) language model’s hidden states. Where n is the model’s hidden dimension, and $z \in \mathcal{Z}$ is the last hidden activation of the forward pass. We wish to write the forward pass at a particular time point as $F : \mathcal{Z} \rightarrow \mathcal{Z}$, and so state it slightly unusually as

$$z_{t+1} = F(W[\text{sample}(\sigma(Vz_t))]).$$

Here $V \in \mathbb{R}^{l \times n}$ and $W \in \mathbb{R}^{l \times n}$ are the un-embedding

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and embedding vocabulary matrices respectively, that take a vector in the hidden activation space to the vocabulary space (or vice-versa), $\mathcal{V} \cong \mathbb{R}^l$, with vocabulary size l . σ is the usual softmax function with a fixed temperature of 1. The $W[\cdot]$ notation is $W[i] = e_i^T W$, so selecting the i -th row from the vocabulary matrix, and *sample* is the function used to select an index from the $\sigma(Vz_t)$ probability distribution.

F depends on the history of the token sequence leading up to a time t , but we will treat this implicitly to lighten notation.

Now, for some generation points, z_t might be oriented in such a way that it is not well aligned to any of the token embedding vectors, and so gives rise to a probability distribution with significant mass on the top two tokens. By analogy to the image generating feed forward network, we say that these points have become *blurred*: The model is producing an “averaged” output between the top two tokens. At these points there is no good choice *sample* can make and — because the input into the model at the next time step is the selected token’s embedding vector — some of the information that z_t was capturing must have been destroyed by the token selection.

Let $v_1 \in \mathcal{Z}$ and $v_2 \in \mathcal{Z}$ be the top and second best token embedding vectors as determined by the sampling process at a time t :

$$\begin{aligned} v_1 &= W[\text{top}(\sigma(Vz_t))], \\ v_2 &= W[\text{second}(\sigma(Vz_t))]. \end{aligned} \tag{1}$$

In this paper, we are going to develop how the volume of the parallelepiped spanned by the vectors z_t , $(p_1 v_1)$, and $(p_2 v_2)$ (where p_i are the probability masses) measures the extent of blurring, and how the wedge product¹ between them leads to an oriented measure of uncertainty. We postulate that the geometry of blurring relates the model’s internal

¹Wedge products are in some sense a generalisation of the cross-product to higher dimension. A wedge between two vectors in three dimensional space $A \wedge B \equiv A \times B$ is exactly the cross-product. Through this, wedge products also generalise the notion of volumes and areas, so that $\|A \wedge B \wedge C\|$ is the volume of the parallelepiped spanned by these three vectors. Note that the ordinary cross-product is only well defined in three dimensions.

state to its learned world model, and show that this product defines one such relationship. We also show that similar but much simpler transformations do not successfully relate the internal state to the learned world model.

2. Nomenclature and Experimental Notes

Before deriving our geometric relationship, let us outline the experimental context we are working in.

All our experiments were performed using the Qwen2.5-32B-Instruct (Qwen Team, 2024) and Mistral-Small-3.1-24B-Instruct-2503 (Mistral AI Team, 2025) models. We developed everything using the Qwen model first, only evaluating the Mistral model after finalizing our claims to reserve it as a held out model.

Beyond the definitions made in the introduction, we will treat $y \in \mathcal{Y} \cong \mathbb{R}^n$ as the vector in the residual stream at layer 20 and 30 of the Qwen and Mistral models, with $\mathcal{Y}$ the vector space of the residual stream. We call these layers the *probe layer* of the Qwen and Mistral models for reasons that will become clear. Due to the models’ use of RMSNorm (Zhang & Sennrich, 2019), the final activations have similar magnitude, and so we make the approximation throughout that they all lie on the S^{n-1} manifold². Therefore z_t can be restricted to $z_t \in S^{n-1}$. Finally, we define the vector $\mu \in T_{z_t}S^{n-1}$ to be in the tangent space of S^{n-1} at z_t , and use the symbol when referring to any such tangent vector.

Throughout we use chess as a controlled and inspect-able reasoning environment to analyse the model’s behaviour. We do this by prompting it with Portable Game Notation (PGN) descriptions of chess positions. We instruct it to think step-by-step and find the best move for the player whose turn it is. We collect a dataset of *significant branch points*, which are generation points where exchanging the top two tokens in otherwise greedy completions changes the final move the model recommends. At such points we also collect last hidden activations, the internal state at the probe layer, and an evaluation of the position (i.e. which side is winning) in centi-pawns using the Stockfish chess engine (The Stockfish developers). By brute forcing this problem, that is checking the output for each possible token swap³, we produced 4, 000 such points. See figure 1 for the exact prompt template and an example branch point.

As mentioned in the introduction, we wish to analyse the learned world model, and so we follow the interpretabil-

²Strictly speaking S^{n-1} is the unit $n - 1$ sphere, so we can either divide all z_t by their constant norm (usually something like $\sqrt{n}$), or redefine the manifold as the $n - 1$ sphere with radius equal to the z_t ’s norm. The choice is arbitrary, but to lighten notation we will make it implicitly and keep speaking of S^{n-1} as the manifold.

³We filter to only consider points where $p_2 > 0.1$ to ease computational load.

ity ideas from OthelloGPT (Li et al., 2023; Nanda et al., 2023; Hazineh et al., 2023) to train 737 linear probes that correspond to binary classification tasks of labels like `mine_knight_f3` or `your_pawn_h4`. They classify $y \in \mathcal{Y}$, which is why we choose the name probe layer for layer 20 and 30 of the models. These particular layers were chosen as they gave us the best probe accuracy.

Using chess to investigate model state tracking has been explored in (Toshniwal et al., 2022). Emergent chess world models with linear probes were also explored in (Karvonen, 2024). Unlike the cited works, we do not train or fine tune the model in any way (extracting linear world representations from off-the-shelf models has also been explored in (Gurnee & Tegmark, 2024)). A difficulty with this is that chess is extremely sparse: Taking a random walk through the tree of legal moves, a pawn, for instance, may only appear on h4 in a vanishing fraction of all positions. For this reason, we generate many legal random games and select a subset that balances the true/false labels for all the probes simultaneously.

Table 1 shows an overview of the accuracies and F1 scores we managed to achieve on the balanced dataset for Qwen and table 2 for Mistral. We note that the probe quality is significantly worse (but still much better than random) for the Mistral model, which will have implications for our analysis later.

Table 1. Probe Performance Metrics (8 Representative Probes, Sorted by Accuracy) Qwen Model

Rank	Probe Label	Accuracy	F1 Score
1	mine_bishop_on_f1	0.825	0.826
...	...	...	...
106	mine_bishop_on_c8	0.922	0.922
211	mine_king_on_e6	0.956	0.957
...	...	...	...
368	mine_knight_on_a2	0.966	0.966
369	mine_knight_on_f2	0.966	0.967
...	...	...	...
474	yours_rook_on_d2	0.969	0.967
632	mine_rook_on_b3	0.978	0.977
...	...	...	...
737	mine_king_on_a8	1.000	1.000

Table 2. Probe Performance Metrics (4 Representative Probes, Sorted by Accuracy) Mistral Model

Rank	Probe Label	Accuracy	F1 Score
1	mine_pawn_on_a4	0.760	0.763
...	...	...	...
368	mine_knight_on_d2	0.887	0.888
369	mine_knight_on_b8	0.887	0.885
...	...	...	...
737	yours_king_on_a4	0.989	0.875

As a corollary, this shows that off-the-shelf instruct tuned

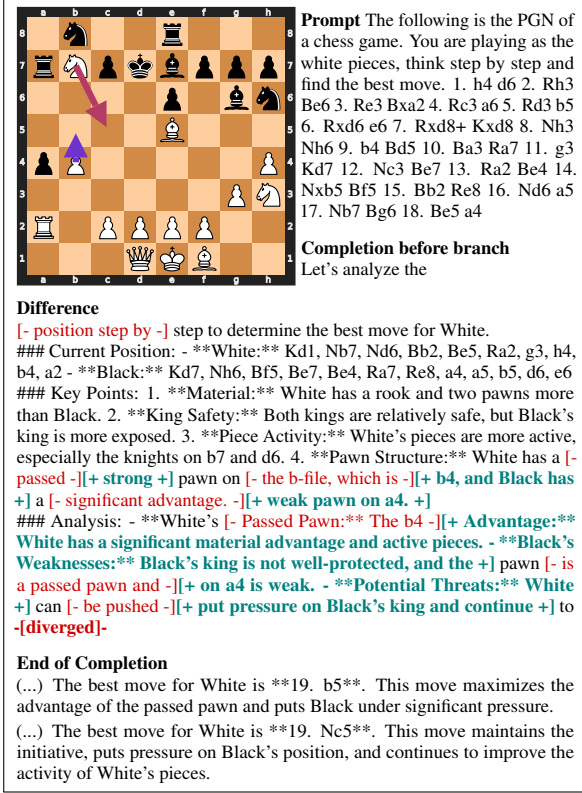


Figure 1. Impact of single-token perturbation at position 4 on reasoning and final move recommendation. Shows the completion before branching and then the difference between the continuations. In the difference, black, red, and green colours indicate coinciding, deleted, and new text in the perturbed completions.

language models implicitly learn a representation of chess. The linear probes have a form of $l(x) = w \cdot x + b$, where w is a vector and b is a scalar. We refer to w as a *world vector*. Prior work on analysing neural network representations is for instance (Zou et al., 2025).

3. Motivation

Prior work (Bhargava et al., 2024; Lu et al., 2022) and our own investigation establishes that language models are significantly unstable to single token changes. Comparing the evaluation of the position after playing the move recommended by the *greedy* and *branch* completions (see section 2 on how these are generated), the mean absolute centi-pawn change is 4.5 ± 1.5 log-centi-pawn in log-space (a full Knight is approximately $\ln(300) \approx 5.7$ log-centi-pawns), highlighting the significance of such changes.

We note that the difference in word choice between the *greedy* and *branch* completions that causes a notably different move to be recommended seems incredibly minor (“analyze the *position*” vs “analyze the *step*” in figure 1), and that the generation streams take many tokens before

they truly begin to diverge in their output (see difference in figure 1).

While sensitivity to small perturbations is expected in auto-regressive systems, explored for instance here (Wang & Zhou, 2024; Holtzman et al., 2020), we take the slightly more radical position that the phenomenon ought not to be dismissed entirely as exponential divergence due to compounding error (see prior work (Zhang et al., 2024)). To that end we postulate that the geometry of blurry points is related to the internal state in a structured, semantically meaningful way; a new kind of relationship that is not captured by considering the exponential divergence alone.

4. Developing the Geometry

We postulate (and then empirically show) that treating

$$A(z_t) = 4z_t \wedge (p_1 v_1) \wedge (p_2 v_2) \quad (2)$$

as giving rise to a geometry that “twists” and turns the internal state y in a precise manner leads to a measure of uncertainty with an intrinsic orientation. This is in contrast to scalar uncertainty measures like in (Kuhn et al., 2023) or the entropy or variance, which do not possess a notion of orientation. Our key results show that this orientation of uncertainty *couples* (i.e. absolute cosine similarity) to the world vectors we’ve extracted from the models’ probe layers. While our construction is more akin to a gauge theory than general relativity, a geometric treatment of language models has also been made in (Kim et al., 2023; Shao et al., 2018).

A point introduces a large error due to sampling when the angle between the predicted z_t and the selected continuation embedding vector is large. We are not concerned with all generation points that have large sampling error: If p_2 is small despite z_t and v_1 not being well aligned, then the error is high, but we have no choice; the second most likely token is too improbable to be sampled without causing artifacts in the model’s output. On the other hand, when v_1 and z_t are not aligned, and p_2 is reasonable, then it is possible to sample the second most likely token without breaking the auto-regressive process F . We are concerned with this second class of points, at which z_t is between two possible continuations which may both reasonably be chosen.

To explain equation 2, consider the parallelepiped spanned by the vectors z_t , $(p_1 v_1)$, and $(p_2 v_2)$ in figure 2. The volume of the parallelepiped depends on the opening angle between all the vectors that span it and these vectors’ lengths. If the vectors are aligned, the volume is small because the parallelepiped is shallow. If $(p_2 v_2)$ is short because of low p_2 probability mass, then the volume is small because one of the object’s faces becomes narrow. Its volume is only great if the vectors aren’t aligned and the total probability mass is somewhat evenly distributed between p_1 and p_2 . This is

exactly the property of the blurry points we want to look at.

The three-blade of equation 2 contains this geometry algebraically. Particularly, the Frobenius norm of the wedge product gives the volume of the parallelepiped, such that

$$\|A(z_t)\|_F \propto \text{extent to which } z_t \text{ is a blurry point.}$$

To develop equation 2 into defining a geometry that affects the internal state, we will upgrade it to a connection by contracting with the tangent vector μ . In general, the connection is an $n \times n$ matrix that captures the infinitesimal change due to the underlying geometry when moving (infinitesimally) along a particular direction. In our case, the underlying geometry is defined by the parallelepiped of our blurring analogy as captured through equation 2, and the direction is the μ tangent vector. Performing the contraction with μ :

$$A_\mu(z_t) = 4p_1p_2(-(\mu \cdot v_1)(z_t \wedge v_2) + (\mu \cdot v_2)(z_t \wedge v_1)). \quad (3)$$

Where we've used that $\mu \cdot z_t = 0$, since μ is in the tangent space at z_t . This expression is a linear combination of two-blades. Using the isomorphism⁴ $\wedge^2(\mathbb{R}^n) \cong \text{Skew}_n(\mathbb{R})$, we notice that each two-blade is an element in the $\mathfrak{so}(n)$ Lie algebra, and so $A_\mu(z_t)$ is a $\mathfrak{so}(n)$ valued 1-form. Said another way, acting with $A_\mu(z_t)$ on a y , that is $A_\mu(z_t)y$, imparts an infinitesimal nudge⁵ corresponding to a rotation defined by the combination of each of the two-blades. Each two-blade (for instance $z_t \wedge v_2$) defines a parallelogram, the same way the three-blade gave a parallelepiped, with the matrix obtained through the isomorphism for the two-blade being a generator of rotations in the plane of the parallelogram (so in the example, a rotation rotating z_t into v_2).

⁴Particularly $\phi(A \wedge B) = BA^T - AB^T$, where ϕ is an isomorphism. See, for instance, (Doran & Lasenby, 2003) for details.

⁵That is, the instantaneous velocity of the rotation. Since $SO(n)$ is a linear Lie group, if you apply the matrix exponential to this generator you are integrating following the velocities, and get a full $SO(n)$ rotation matrix corresponding to the generator back.

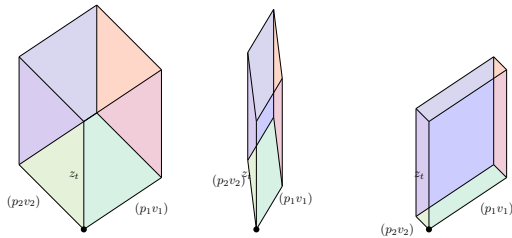


Figure 2. Parallelepiped spanned by z_t , $(p_1 v_1)$, and $(p_2 v_2)$ in three cases (left to right): Vectors are not aligned and probability mass is evenly distributed, probability mass is evenly distributed but vectors are aligned, vectors are not aligned but $p_2 \ll p_1$.

To formally treat equation 3 as a connection, we must attach a copy of $\mathcal{Y}$ to every point on the S^{n-1} output manifold, with equation 3 acting as the connection between the independent vector spaces in this fibre bundle.

4.1. Parallel Transport

Equation 3 is the infinitesimal effect on y due to the blurring geometry. The total change on y due to blurring along a curve γ on the S^{n-1} manifold is given by the parallel transport. This is exactly how parallel transporting a state vector in the $U(1)$ fiber bundle of electromagnetism with the vector potential as the connection tells us how the electromagnetic field changes the internal phase of the quantum mechanical system.

The parallel transport operator along a curve defined by infinitesimal segments δx_i is given by the repeated application of the matrix exponential of our connection

$$U = \lim_{N \rightarrow \infty} \prod_i^N \exp(-A_{\delta x_i}(z_i)), \quad (4)$$

where each z_i is given by the recursion $z_{i+1} = z_i + \delta x_i$. $A_\mu(z_t)$ is in principal smooth and differentiable for any z_t , as long as $A_\mu(z_t + \delta)$ does not change which two tokens are the top ones. This means (if one does not want to apply further tricks) the path γ must be contained to a region around z_t where the top two tokens are preserved.

Formally taking the limit in equation 4 leads to the parallel transport operator for a curve γ as the path ordered integral operator

$$U = P \exp \left(- \int_0^1 A_{\dot{\gamma}(s)}(\gamma(s)) ds \right). \quad (5)$$

With P the path ordering symbol⁶ and where we explicitly parameterised the curve.

Given a path on the output manifold, we can now apply U to a y and see how the state vector changes due to the blurring geometry along the path.

5. Probability Charge

We would like to specifically highlight the $4p_1p_2$ product in our expression for $A_\mu(z_t)$. It mediates the coupling strength of our connection, and so, in analogy to the charge prefactor in the connection of electromagnetism, we call it the *probability charge*.

⁶Since each $A_{\dot{\gamma}(s)}(\gamma(s))ds$ in equation 5 is a matrix we must be careful to maintain multiplication order. The path ordering symbol indicates that the exponentiated integral must resolve back to equation 4 instead of something like $\exp(a + b + \dots)$.

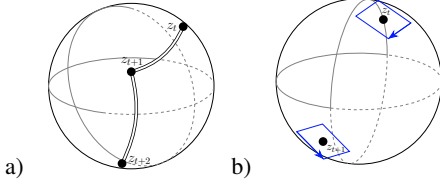


Figure 3. (a) The manifold of the last hidden activations produced by the F process. Each activation lies on the surface of an (ambient) sphere, and is connected via an instantaneous tunnel that it takes no time to travel through. (b) Two holonomy operators at two different points on the last hidden activation sphere.

In the same way that a chargeless particle does not experience the electromagnetic force due to that pre-factor being zero, an internal state does not experience the blurring geometry at a confident generation point when all the probability mass is concentrated in p_1 , due to p_2 being zero. If we for a moment assume all the probability mass is concentrated between the top two tokens, then we may write $p_2 \approx 1 - p_1$, and the probability charge becomes $4p_1(1 - p_1)$. This is an inverted parabola with a maximum of 1 when $p_1 = 0.5$ — a state experiences the blurring geometry maximally at maximum uncertainty.

6. Holonomy

We cannot take a state vector and parallel transport it from a z_t to a z_{t+1} in a meaningful way, as such a transport would require choosing a path which passes through many output states that the model never produced. We also cannot deform the output manifold to construct tunnels between z_t and z_{t+1} (see figure 3a), since we still need to choose a path through the ambient space that will ultimately traverse invalid intermediate states. In fact, with such a construction, the curvature experienced through the tunnel is exactly the accumulated curvature of the path we choose to construct it and so we wouldn't have resolved our path-dependence anyways.

We can, however, parallel transport a state vector y along a closed loop around a particular z_t , measuring only the local geometric effects. This is the holonomy operator, and it measures the extent to which closed loops on the manifold fail to close in the attached internal space due to the underlying geometry. We wish to use this to find physical effects due to the blurring geometry on the state vector y . One could construct such operators for each z_t (see figure 3b) and apply them in time-ordered fashion to get the total holonomy due to a sequence up to time t

$$H_{\text{total}} = H_t H_{t-1} \dots H_1.$$

To compute H for a particular z_t , we must choose a closed loop around z_t . Let us choose a plane at z_t spanned by

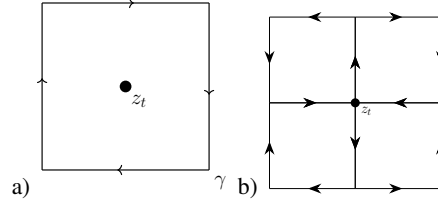


Figure 4. (a) The path around z_t we parallel transport along to find the holonomy. (b) A clover formed from four squares in the $u - v$ plane centred on a point z_t to avoid artifacts due to the local coordinate choice.

$u, v \in T_{z_t} S^{n-1}$, and construct the path in this plane. We wish to simply go around z_t in a square with some small side length. However, choosing the obvious path (figure 4a) to do this introduces artifacts at order ϵ^3 due to the local coordinate choice. A standard construction (Gattringer & Lang, 2010; Baez & Muniain, 1994) to resolve this is to compute an averaged clover like shown in figure 4b of four small squares with side length ϵ . This yields $\mathcal{O}(\epsilon^4)$ accuracy.

Notice that if ϵ is small enough, all points on the path defined by these squares are still approximately in the tangent space of S^{n-1} at z_t ⁷. By this construction, the holonomy at a point z_t is

$$H_{z_t} = \frac{1}{4}(H_{z_t}^{(1)} + H_{z_t}^{(2)} + H_{z_t}^{(3)} + H_{z_t}^{(4)}),$$

where the upper index refers to the different squares in the clover. Each such square's holonomy is the path-ordered product of the parallel transport around its legs. To find each leg we must apply equation 5 four times with the path always being a straight line in the u or v direction. Since ϵ is small, we can Taylor expand the integrand in $-\int_0^1 A_{\dot{\gamma}(s)}(\gamma(s))ds$ in powers of ϵ . As an example, for the top-right rectangle in the clover this leads to the legs

$$\begin{aligned} L_1 &\approx \epsilon A_\mu(z), & L_2 &\approx \epsilon A_\nu(z + \epsilon\mu), \\ L_3 &\approx -\epsilon A_\mu(z + \epsilon\nu), & L_4 &\approx -\epsilon A_\nu(z). \end{aligned}$$

where we've dropped all $\mathcal{O}(\epsilon^2)$ terms here. Using these, the holonomy operator for the top-right square in the clover then is

$$H^{(1)} = e^{-L_4} e^{-L_3} e^{-L_2} e^{-L_1}.$$

Now we can apply Baker-Campbell-Hausdorff⁸ for each of the matrix exponentials, collect in powers of ϵ , and sum the holonomy contributions from each square in the clover to

⁷This is why the path for the squares in the clover does not need to bend along the sphere's surface.

⁸This formula gives how to combine products of exponentials into sums in the exponent for objects that don't commute. To first order it is $e^x e^y \approx e^{x+y+1/2[x,y]+\dots}$

arrive⁹ at the final holonomy operator at a point z_t

$$H = \exp(h)$$

$$h \approx -\epsilon^2(\partial_\mu A_\nu(z) - \partial_\nu A_\mu(z) - [A_\nu(z), A_\mu(z)]) + \mathcal{O}(\epsilon^4), \quad (6)$$

where this being accurate to order ϵ^4 despite only containing terms up to order ϵ^2 is a consequence of the clover construction. Also note that we can recognise the term involving the partial derivatives and the matrix commutator¹⁰ as the standard curvature tensor and that there is a global sign freedom introduced by choosing clockwise or counter-clockwise paths for the clover¹¹.

At each z_t there are n choose 2 many planes tangent to the S^{n-1} manifold. To avoid having to compute all of them, we only consider the plane with the largest magnitude contribution. We construct this by projecting v_1 onto the tangent space of z_t , forming u , and constructing v as the projection of v_2 onto the tangent space at z_t and orthogonalising to u .

The curvature that is captured as h of our blurring geometry is a “physically” meaningful observable, which we will now test.

7. Experimental Design

In section 2 we’ve already outlined our experimental context and have described how we have access to linear probes for the chess world model. We’ve also explained how we’ve generated the *greedy* and *branch* continuations.

At each *branch point* we can compute a holonomy operator according to equation 6. We also have access to two y values: The first is the activation in the residual stream at the probe layer for the transition from the branch point to the next greedy token, and the second is the same but for the transition from the branch point to the branch token. We call these the *greedy y* and *branch y* respectively.

Since we have access to the true board state we prompted the model with, at all points we can choose to only look at the probes which are currently on the board or not. We call these the *active* and *bulk* probes.

Let us define the difference of applying H and not as $q = Hy - y$ and note that $q = Hy - y = (H - \mathbb{1})y \approx hy$, since $H = \exp(h) \approx 1 + h + \dots$. We can recognize h and consequently $H - \mathbb{1}$ (approximately¹²) as the curvature of

⁹The authors implemented this symbolically using SymPy (Meurer et al., 2017) to generate a JAX (Bradbury et al., 2018) function for H_{z_t}

¹⁰This is $[A, B] = AB - BA$

¹¹We choose counter-clockwise throughout.

¹²Since h is $\mathcal{O}(\epsilon^2)$, keeping only the linear term in the Taylor series of the matrix exponential is $\mathcal{O}(\epsilon^4)$ accurate. Extending the Taylor series to the square order term would only introduce a $\mathcal{O}(\epsilon^4)$

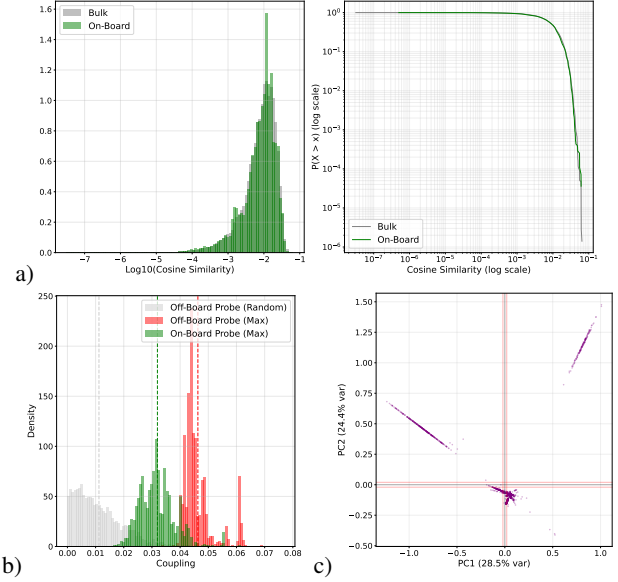


Figure 5. (a) Log space density and log-log survival plot of bulk and active distributions coupling strengths, showing they are inseparable. (b) Distribution of coupling strengths to random world vector from the bulk (gray), maximum probe from the active set (green), and maximum probe from the bulk set (red). (c) Two dimensional PCA scatter of maximal probe couplings.

the blurring geometry.

q is the vector one gets after applying the curvature due to blurring to y . One can simply treat it as an “uncertainty vector” if one so wishes, similar to how one can treat the entropy or variance as an “uncertainty scalar,” without invoking more of the underlying mathematics. Notice that in a well behaved theory we expect only the curvature to give us physical results — the holonomy on its own is not an observable.

We call the absolute cosine similarity the *coupling* between q and the world vectors w_i . We want to try to separate the probe distribution in some way. The first attempt is to simply consider the active and bulk set separately, but figure 5a shows that they cannot be separated in this way. We can separate them however by considering the bulk and active set separately, and selecting the maximum coupling probe from these two sets (see figure 5b). We will use the maximum coupling probe throughout.

Since we are going to analyse the q vectors, we are only using h from equation 6, and so one might be inclined to not invoke the full machinery of section 6. This would mean forming the connection of equation 3 and then investigating the implied non-abelian curvature, which is exactly h at $\mathcal{O}(\epsilon^2)$, directly. If you do this, you would have to insert a to-be-determined pre-factor, like our ϵ , into the curvature, correction.

since not writing it is just having it implicitly be 1. You’d then lose the clean interpretation of the pre-factor as the side length of the clover and you would need to use some empirical procedure to determine its value. Particularly that last part would introduce issues of (over)-fitting the expression to whatever downstream task you use for the empirical procedure, which we have avoided. Further, omitting the construction of the holonomy operator loses the information of which other terms exist in h : we only know that there is exactly an $\mathcal{O}(\epsilon^2)$ term and then further $\mathcal{O}(\epsilon^4)$ terms because of deriving h from the full holonomy operator. If we had guessed to analyse the curvature directly, we’d need to use some empirical method to exclude other terms contributing, which would again fit the expression to the downstream task. With the holonomy construction, such other terms would simply show up at different orders. We can also compute all higher order terms, and know exactly how they depend on the clover construction. The effects of higher order terms is something to analyse in future work¹³.

8. Results

8.1. Principal Component Analysis and Chess Board Files

We ran two dimensional PCA on the population of q vectors to see if there is semantic structure to them. This would indicate that the geometry of blurring derived purely from the token space geometry is semantically meaningful in an earlier layer, revealing a relationship between the blurring failure mode and the model’s internal structure. Looking at figure 5c for a scattering of the PCA results, we notice two “ears” of points in the top left and top right quadrants, away from the bulk in the centre. Selecting out the right and left ears and computing the coupling to the maximal probe, we

¹³It is in principal possible to search for geometries where $\epsilon > 1$ is possible. At these points, if they exist, higher order terms are not suppressed at all, and a completely different analysis would need to be made.

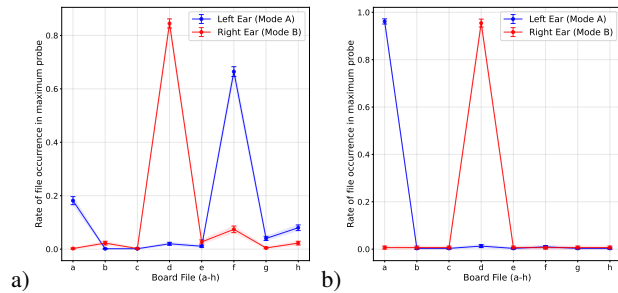


Figure 6. Distribution of chess file occurrences in maximum probes for left and right ear distributions for the greedy internal state (a) and branch internal state (b). Error is estimated using Bayesian methods by assuming a flat prior.

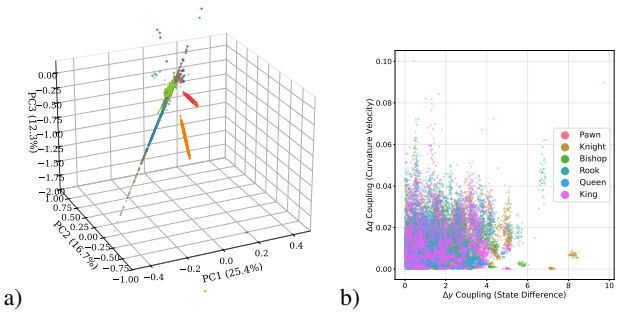


Figure 7. a) Scattering of coupling strengths as three dimensional PCA. Shades of blue and green indicate points belonging to left and right ear identified in two dimensional PCA. Red and yellow lines indicate greedy and branch points that form similar but separate lines. b) Coupling of Δy to probes (by piece type) vs coupling of Δq to same probes. Shows that the blurring geometry still twists the world model between the two possible continuations, even if the direct difference between the two branches’ internal states is orthogonal to the world model.

can aggregate the result by which file (on the chess board a to h) the maximal probe feature belongs to. See figure 6a and b for the rate at which each file occurs for the greedy and branch ys . For both, we notice that they are separable distributions, with one ear peaking on the d file and the other on the f or a files. It appears that the rotation induced by H is not random, but instead seems to separate king’s & queen’s side and centre world vectors. Notice also that the scattering induced by H for ys from the greedy continuation seems to mirror the semantic meaning of the one induced on the ys from the branch continuation. This appears to be a property of the holonomy operator directly, as it is the same in both cases, derived from the blurring geometry at the branch point, with only the y it is being applied to changing.

8.2. Three Dimensional PCA

We can extend this approach straightforwardly to three dimensions. This reveals a new set of interesting points (see figure 7a) formed by two lines which are not classified by our existing ears structure, and separate themselves into one for the greedy and one for the branch continuations. That is, they are q vectors where all the ones from a branch continuation lie in a line, alongside but entirely separate from a line formed by the greedy continuation q vectors. We can use this as an opportunity to analyse the difference in the q vectors between greedy and branch continuations.

First, we compute the average coupling strength of the qs to all the probes for both the greedy and branch ys . We scatter these in figure 8a. We notice a clear trend: The coupling in the branch continuation is systematically stronger than in the greedy completion. Again, we’ve applied the same H computed with the same arguments to the different ys . It appears that the internal state of the branch path is systemat-

ically more entangled with the blurring curvature than that of the greedy branch, where we are careful to note that there is a global sign freedom¹⁴.

Let q' be the q vector of the branch path, while q is this vector for the greedy path. We define $\Delta q = q' - q$, which measures the flow induced between the two internal states across the branch boundary. Figure 8b shows the coupling strength of Δq with the probes sorted by *mine/yours* and the piece type. We notice that the coupling strength appears to follow the general importance of chess pieces (Spearman’s¹⁵ $\rho = 0.90$, $p = 0.037$ for the *mine* pieces). With the coupling being almost entirely orthogonal to pawns, a little stronger for knights, then bishops, and finally strongest for the two most powerful pieces.

This means the difference in the rotation induced by H for the two possible y s we apply it to (we stress that H is the exact same matrix in both cases) couples to world vectors at a strength monotonically related to the relative chess piece importance.

8.3. Internal Model Shift vs Holonomy Shift

We can scatter the coupling of Δq defined in the previous section vs the absolute shift in internal state $\Delta y = y' - y$.

Figure 7b shows just this scatter. Notably, there is no clear trend line, and even for points on the left where Δy is small, we see many points where Δq is large. This means that despite the internal state shifting between the greedy and branch continuations in a manner that is orthogonal to the chess world model, our oriented measure of uncertainty does

¹⁴Which means that if we were to choose a clockwise oriented clover in the construction, then the greedy path would be more entangled than the branch path. The notable result is that there is a stark preference for one or the other, but not which one.

¹⁵Computed by correlating coupling strength with standard piece value in pawns. We used 3.5 pawns for the bishop and exclude the king.

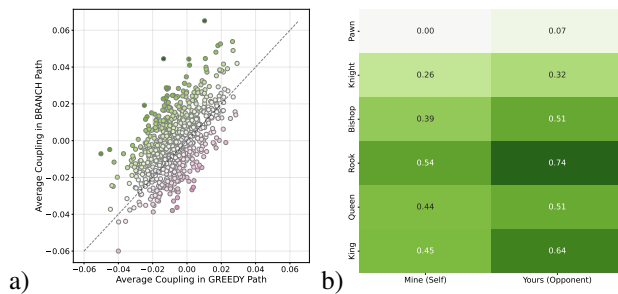


Figure 8. (a) Average probe coupling to q vectors for the greedy and branch continuation plotted against each other. We note the systematically stronger coupling in the branch continuation. (b) Average coupling strength of Δq vector to probes by piece type and opponent / own piece classification.

still rotate along the world model. We find this a remarkable result, but are unable to offer further analysis here.

8.4. Ablations

To verify that the observed structure arises from the specific geometry of token-space blurring rather than arbitrary rotations, we applied a random $SO(n)$ matrix matching the Frobenius norm of the real H operators to the same internal states. The resulting PCA shows no directional structure (see appendix figure 10) and the two principal directions capture a lower fraction of the dataset variance (53.9% vs 34.8%), confirming that the semantic clustering is specific to our derived holonomy operator.

As a further ablation (see appendix figure 10) we test applying an operator that rotates z_t onto v_2 and one that rotates z_t onto v_1 . Similarly, we do not see a structure forming in the PCA and only 31.4% of the total variance is captured in the two principal directions. It has been shown (nostalgebraist, 2020; Belrose et al., 2025) that one can get semantically meaningful logit distributions by applying V to intermediate layers. That a simple rotation like this does not give us an interesting PCA then is strong evidence that it is the tension between competing output vectors (i.e. blurring) and our particular operator form that is driving the structure we’re observing.

Computing the coupling for Hy instead of $q = Hy - y$ didn’t lead to any semantically interesting results. As mentioned, the curvature of the field is the physical observable and so it is a sign that things are well behaved that we saw semantic coupling only for the q population (Recall that $Hy - y \approx hy$, where we recognise h as the curvature in equation 6).

8.5. Mistral

As mentioned in section 2, we’ve reserved the Mistral model to analyse generalisability of our results.

See appendix figure 9 for a collection of the associated graphs. We note that, similar to the Qwen model, the probe couplings can be separated through the maximum *active* and *bulk* probes, and that the magnitude of the coupling strength is of similar order. Similarly again, running two dimensional PCA leads to clear separable lines. We notice that the magnitudes of the PCA1 and PCA2 axes are much smaller for the Mistral model than for the Qwen model. We believe that this is due to the poorer probe quality mentioned in section 2, causing the data to be more noisy. Because of this small magnitude, we’ve multiplied the coupling values by 400 in the semantic shift and spectrum for the visualisation only. This doesn’t affect Spearman’s ρ or the relative effect.

Notably, the three dimensional PCA also contains lines of

separated *greedy* and *branch* points. Interestingly the flow for the Mistral model seems to be from the *branch* to the *greedy* states, opposite what we’ve observed for the Qwen model.

As mentioned in section 6, there is a global sign freedom in our construction, and so when looking at a single model we can only analyse the magnitude and consistency of the flow between *greedy* and *branch* states, not its direction. That being said, if we pick the same convention between two models, then the difference in the flow direction *between the models* is an interpretable prediction of our framework. We made the same choice for the Qwen and Mistral models and so the difference must be attributed to a genuine difference between the models: potentially something about the training stack or slight architectural choices might determine the orientation. A precise analyses of the origin of this difference in future work is likely very interesting, especially because of the consistency in this flow as seen in figure 8a and appendix figure 9.

Isolating these lines, we can run the same semantic shift analysis as before. We note that we see a similar relationship between piece importance and coupling strength ($\rho = -0.90$, $p = 0.037$ for the `mine` pieces), except that the King appears to be an outlier (for Qwen and Mistral the piece was not included as it cannot be captured, and so no pawn value is commonly assigned to it). The spectrum result shows less structure than in the Qwen case (we didn’t analyse this structure in either case), but does show the same interesting points where Δy is small but Δq is not.

Overall, the Mistral data confirms that our results are not a quirk of the Qwen model and that geometric treatment of blurring seems to generalise across model lineages. It is striking how similar the principal component structure looks in both cases.

9. Conclusion

We have introduced a novel geometric framework for understanding the relationship between token sampling ambiguity and language model internal states. Unlike scalar measures of uncertainty like entropy or variance, our framework provides an oriented quantity. We’ve shown that this orientation couples directly to the model’s learned world model in ways related to the strategic nature of our reasoning task.

Our central contribution is that this measure is derived exclusively from the geometry of the token embeddings and yet successfully interrogates the complicated representation these general purpose LLMs learn. It implies that the indecision between possible next tokens is not a statistical fluke, while being a (of possibly many) precise — structurally rich — relationship between the output geometry and the model’s internal representation.

We hope this geometric perspective contributes to a deeper understanding of how the discrete, combinatorial nature of language interfaces with the continuous geometry of neural networks, and believe this could lead to a more mechanistic understanding of the representations neural networks learn.

Our analysis is limited by being restricted to a single problem domain and only two general purpose models. Future research can attempt to relate the blurring geometry to reasoning tasks with harder to define strategic nature.

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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Appendix A

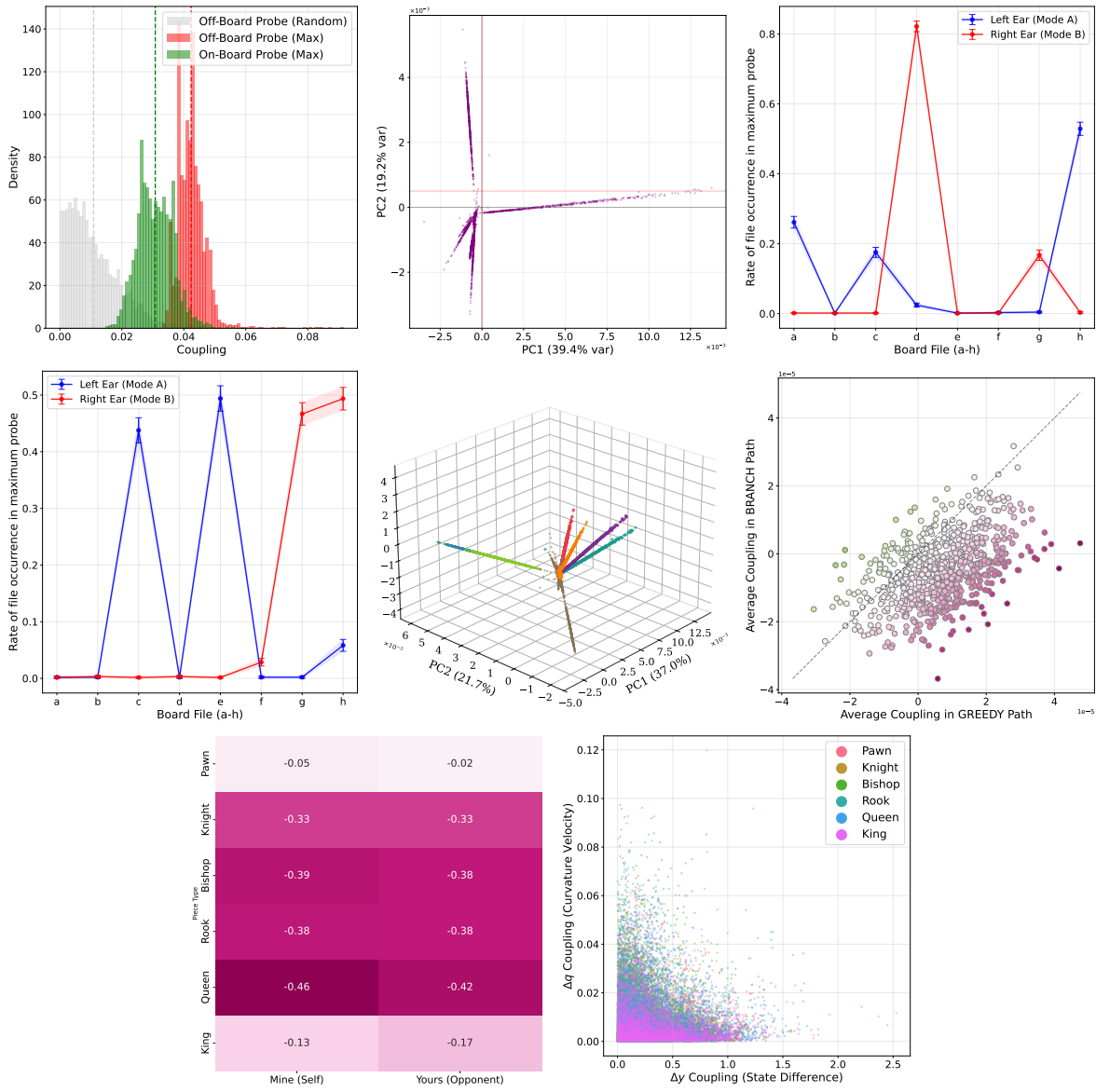


Figure 9. Collection of results graph like those in figure 5, 6, 8, and 7 for the Mistral model.

Appendix B

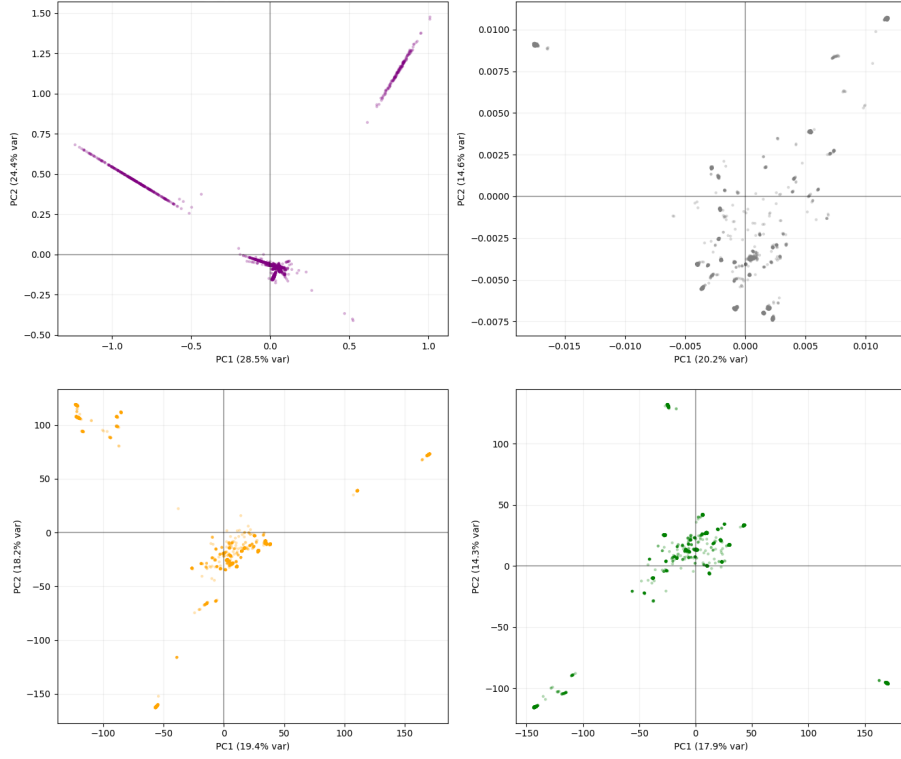


Figure 10. PCA projections of $q = Hy - y$ for the true holonomy operator, versus a scale-matched random $SO(n)$ matrix, and a holonomy operator rotating z_t onto v_1 and one onto v_2 . The structured V-pattern and higher variance concentration (52.9% vs 34.5% and 31.4%) in the true holonomy confirms that the semantic clustering arises from the specific geometry of token-space blurring, not arbitrary rotations.